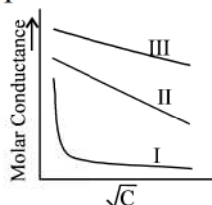


WORK SHEET - 05

ELECTROCHEMISTRY

Single correct:

- Q.1 A graph was plotted between the molar conductance of various electrolytes (HCl, KCl and CH₃COOH) and root of their concentrations in mole per litre.



Which of the following is correct match?

- (A) I (CH₃COOH) ; II (KCl) ; III (HCl) (B) I (HCl) ; II (KCl) ; III (CH₃COOH)
(C) I (CH₃COOH) ; II (HCl) ; III (KCl) (D) I (KCl) ; II (CH₃COOH) ; III (HCl)
- Q.2 Equivalent conductance of 0.1 M HA (weak acid) solution is 10 Scm²equivalent⁻¹ and that at infinite dilution is 200 Scm²equivalent⁻¹ Hence pH of HA solution is
(A) 1.3 (B) 1.7 (C) 2.3 (D) 3.7
- Q.3 A silver wire dipped in 0.1 M HCl solution saturated with AgCl develops oxidation potential of -0.25 V. If $E^\circ_{\text{Ag}/\text{Ag}^+} = -0.799 \text{ V}$, the K_{sp} of AgCl in pure water will be
(A) 2.95×10^{-11} (B) 5.1×10^{-11} (C) 3.95×10^{-11} (D) 1.95×10^{-11}
- Q.4 For the cell : Pt | H₂(0.4 atm) | H⁺(pH=1) || H⁺(pH=2) | H₂ (0.1 atm) | Pt
The measured potential at 25°C is
(A) -0.1 V (B) -0.5 (C) -0.041 (D) none
- Q.5 EMF of a cell is given by $E = (-1.05 + 10^{-4} T^2) \text{ V}$, where T is temperature in Kelvin. Which of the following options are **correct** w.r.t. the galvanic cell.
(A) $\Delta H^\circ_{\text{Rxn}}$ of the cell reaction will be temperature independent.
(B) The cell reaction involves increase in randomness.
(C) At all the temperatures, the cell reaction will be spontaneous.
(D) The cell reaction will be non-spontaneous at T = 200K.
- Q.6 A saturated solution in AgA ($K_{\text{sp}} = 3 \times 10^{-14}$) and AgB ($K_{\text{sp}} = 1 \times 10^{-14}$) has conductivity of $375 \times 10^{-10} \text{ Scm}^{-1}$ and limiting molar conductivity of Ag⁺ and A⁻ are 60 Scm² mol⁻¹ and 80 Scm² mol⁻¹ respectively then what will be the limiting molar conductivity of B⁻ (in Scm² mol⁻¹)
(A) 150 (B) 180 (C) 190 (D) 270
- Q.7 The conductivity of a saturated solution of Ag₃PO₄ is $9 \times 10^{-6} \text{ S m}^{-1}$ and its equivalent conductivity is $1.50 \times 10^{-4} \text{ S m}^2 \text{ equivalent}^{-1}$. The K_{sp} of Ag₃PO₄ is
(A) 4.32×10^{-18} (B) 1.8×10^{-9} (C) 8.64×10^{-13} (D) None of these
- Q.8 Na-Amalgam is prepared by electrolysis of aq. NaCl solution using 10 gm Hg cathode. How many Faraday's of electricity is required to prepare 18.7 % Na-Amalgam. (current Eff. = 50%)
(A) 0.081 F (B) 0.16 F (C) 0.2 F (D) 0.1 F

Assertion and Reason :

- Q.9 **Statement-1:** Electrode potential of any electrode will change on changing any of its intensive properties
Statement-2: Any intensive property of system changes on changing any of its properties whether intensive or extensive.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

Comprehension :**Paragraph for Question Nos. 10 & 11**

Copper reduces NO_3^- into NO and NO_2 depending upon conc. of HNO_3 in solution. Assuming $[\text{Cu}^{2+}] = 0.1 \text{ M}$, and $P_{\text{NO}} = P_{\text{NO}_2} = 10^{-3} \text{ atm}$ and using given data answer the following questions:

$$E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ volt}$$

$$E^\circ_{\text{NO}_3^-/\text{NO}} = +0.96 \text{ volt}$$

$$E^\circ_{\text{NO}_3^-/\text{NO}_2} = +0.79 \text{ volt}$$

$$\text{at } 298 \text{ K } \frac{RT}{F} (2.303) = 0.06 \text{ volt}$$

- Q.10 E_{Cell} for reduction of $\text{NO}_3^- \longrightarrow \text{NO}$ by Cu(s) , when $[\text{HNO}_3] = 1 \text{ M}$ is [At $T = 298$]
 (A) ~ 0.61 (B) ~ 0.71 (C) ~ 0.51 (D) ~ 0.81
- Q.11 At what HNO_3 concentration thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same?
 (A) $10^{1.23} \text{ M}$ (B) $10^{0.56} \text{ M}$ (C) $10^{0.66} \text{ M}$ (D) $10^{0.12} \text{ M}$

Paragraph for question nos. 12 to 14

0.1 mole AgNO_3 is added in 250 ml of saturated solution of AgCl at 25°C without changing volume.

Given : K_{sp} of $\text{AgCl} = 1.0 \times 10^{-10} \text{ M}^2$

Ionic conductance of Ag^+ ion = $60 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

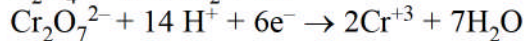
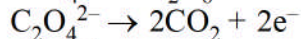
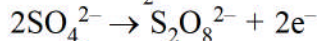
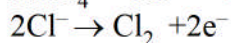
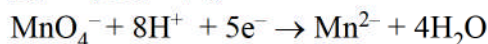
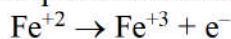
Ionic conductance of Cl^- ion = $75 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

Ionic conductance of NO_3^- ion = $75 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

- Q.12 $[\text{Cl}^-]$ in the final solution is equal to
 (A) 10^{-9} M (B) $2.5 \times 10^{-10} \text{ M}$ (C) 10^{-5} M (D) $2.5 \times 10^{-7} \text{ M}$
- Q.13 If the solution is electrolysed using inert electrodes, the expected electrode products are :
 (A) Ag at cathode and Cl_2 at anode (B) Ag at cathode and O_2 at anode
 (C) H_2 at cathode and Cl_2 at anode (D) H_2 at cathode and O_2 at anode
- Q.14 The conductivity of solution is :
 (A) $5.4 \times 10^{-2} \Omega^{-1} \text{ cm}^{-1}$ (B) $5.2 \times 10^{-2} \Omega^{-1} \text{ cm}^{-1}$
 (C) $4.0 \times 10^{-2} \Omega^{-1} \text{ cm}^{-1}$ (D) $8.4 \times 10^{-2} \Omega^{-1} \text{ cm}^{-1}$

Paragraph for question nos. 15 to 16

In quantitative estimation of any element using an oxidizing agent it is essential to predict which substance(s) gets oxidized by the oxidizing agent. In case more than one substance are getting oxidized then the oxidizing agents gets distributed in all the reactions taking place. From this information and the data given below answer the questions that follow.



$$E^\circ_{\text{O.P.}} = -0.77\text{V}$$

$$E^\circ_{\text{R.P.}} = 1.51\text{V}$$

$$E^\circ_{\text{O.P.}} = -1.36\text{V}$$

$$E^\circ_{\text{O.P.}} = -2.0\text{V}$$

$$E^\circ_{\text{O.P.}} = -0.77\text{V}$$

$$E^\circ_{\text{R.P.}} = 1.33\text{V}$$

- Q.15 Which of the following cannot be used for quantitative estimation of FeC_2O_4 ?
 (A) $\text{K}_2\text{Cr}_2\text{O}_7$ in presence of dil H_2SO_4 . (B) $\text{K}_2\text{Cr}_2\text{O}_7$ in presence of dil. HCl.
 (C) KMnO_4 in presence of dil. HCl. (D) KMnO_4 in presence of dil. H_2SO_4



- Q.16 Millimoles of FeC_2O_4 in the solution if 50ml of 0.1M KMnO_4 is used for its oxidation in presence of dil. HCl if 3.5 millimoles of Cl_2 is obtained along with other products.
 (A) 25/3 millimoles (B) 6 millimoles
 (C) 12.5 millimoles (D) quantitative estimation can't be made from given data.

Paragraph for Question Nos. 17 to 19

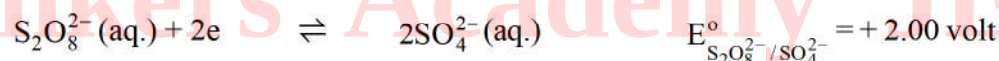
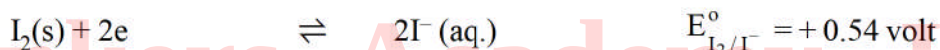
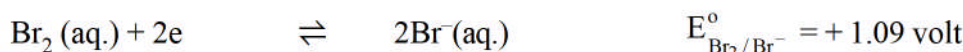
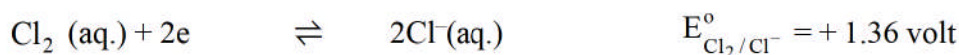
A sample of water from a large swimming pool has a resistance of $10000\ \Omega$ at 25°C when placed in a certain conductance cell. When filled with 0.02 M KCl solution, the cell has a resistance of $100\ \Omega$ at 25°C . 585 gm of NaCl were dissolved in the pool, which was thoroughly stirred. A sample of this solution gave a resistance of $8000\ \Omega$.

[Given : Molar conductance of NaCl at that concentration is $125\ \Omega^{-1}\text{cm}^2\text{mol}^{-1}$ and molar conductivity of KCl at 0.02 M is $200\ \Omega^{-1}\text{cm}^2\text{mol}^{-1}$.]

- Q.17 Cell constant (in cm^{-1}) of conductance cell is:
 (A) 4 (B) 0.4 (C) 4×10^{-2} (D) 4×10^{-5}
- Q.18 Conductivity (Scm^{-1}) of H_2O is:
 (A) 4×10^{-2} (B) 4×10^{-3} (C) 4×10^{-5} (D) None of these
- Q.19 Volume (in Litres) of water in the pool is:
 (A) 1.25×10^5 (B) 1250 (C) 12500 (D) None of these

More than one correct :

- Q.20 Pick out the **correct** statements among the following from inspection of standard reduction potentials (Assume standard state conditions).



- (A) Cl_2 can oxidise SO_4^{2-} from solution
 (B) Cl_2 can oxidise Br^- and I^- from aqueous solution
 (C) $\text{S}_2\text{O}_8^{2-}$ can oxidise Cl^- , Br^- and I^- from their aqueous solutions
 (D) $\text{S}_2\text{O}_8^{2-}$ is added slowly, Br^- can be reduce in presence of Cl^-

Match the column :

Q.21 **Column I**

- (A) Dilute solution of HCl
 (B) Dilute solution of NaCl
 (C) Concentrated solution of NaCl
 (D) AgNO_3 solution

Column II

(Electrolysis product using inert electrode)

- (P) O_2 evolved at anode
 (Q) H_2 evolved at cathode
 (R) Cl_2 evolved at anode
 (S) Ag deposition at cathode

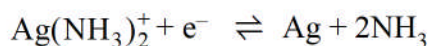
Subjective:

- Q.22 Calculate the potential of an indicator electrode versus the standard hydrogen electrode, which originally contained 0.1M MnO_4^- and 0.8M H^+ and which was treated with 90% of the Fe^{2+} necessary to reduce all the MnO_4^- to Mn^{2+} .



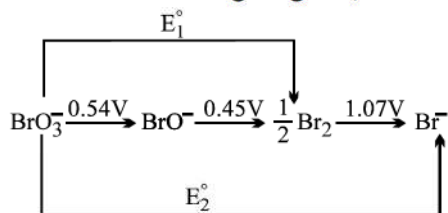
- Q.23 One of the methods of preparation of per disulphuric acid, $\text{H}_2\text{S}_2\text{O}_8$, involve electrolytic oxidation of H_2SO_4 at anode ($2\text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_8 + 2\text{H}^+ + 2\text{e}^-$) with oxygen and hydrogen as by-products. In such an electrolysis, 9.722 L of H_2 and 2.35 L of O_2 were generated at STP. What is the weight of $\text{H}_2\text{S}_2\text{O}_8$ formed?

Q.24 Calculate E° for the following reactions at 298 K,



Given : $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.7991 \text{ V}$, $K_{\text{Ins}}[\text{Ag}(\text{NH}_3)_2^+] = 6.02 \times 10^{-8}$ and $K_{\text{Ins}}[\text{Ag}(\text{CN})_2^-] = 1.995 \times 10^{-19}$

Q.25 From the standard potentials shown in the following diagram, calculate the potentials E_1° and E_2° .

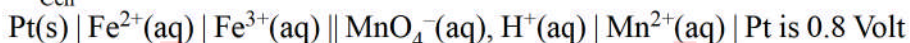


Q.26 If electrolysis of AgNO_3 solution is carried out with 100 gm impure Ag as anode of 90% purity and pure Ag of mass 100gm as cathode by passing 2amp current for 9650 sec. Then the mass of cathode and anode will be :

Q.27 Find $\wedge_m^\infty$ (in $\Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$) for SrCl_2 in water at 25° from the following data.

Conc. C (mole / L)	0.25	1
$\wedge_m$ ($\Omega^{-1} \text{ cm}^2 / \text{mol}$)	260	250

Q.28 E°_{Cell} for the cell :



Given : $\Delta G_f^\circ, \text{H}_2\text{O}(l) = -230 \text{ kJmol}^{-1}$
 $\Delta G_f^\circ, \text{Fe}^{3+}(\text{aq}) = -10.2 \text{ kJmol}^{-1}$
 $\Delta G_f^\circ, \text{Mn}^{2+}(\text{aq}) = -229 \text{ kJmol}^{-1}$
 $\Delta G_f^\circ, \text{Fe}^{2+}(\text{aq}) = -84 \text{ kJmol}^{-1}$

Determine magnitude of $\Delta G_f^\circ, \text{MnO}_4^-(\text{aq})$ in kJmol^{-1} .

Q.29 EMF of the following cell is 0.634 volt at 298 K.



Calculate pH of the anode compartment.

Given : $E^\circ_{\text{Cl}^-|\text{Hg}_2\text{Cl}_2|\text{Hg}} = 0.28 \text{ V}$ and $\frac{2.303 RT}{F} = 0.059$.

Q.30 For the reaction, $4\text{Al}(\text{s}) + 3\text{O}_2(\text{g}) + 6\text{H}_2\text{O} + 4 \text{OH}^- \rightleftharpoons 4 [\text{Al}(\text{OH})_4^-]$; $E^\circ_{\text{cell}} = 2.73 \text{ V}$. If $\Delta G_f^\circ(\text{OH}^-) = -157 \text{ kJ mol}^{-1}$ and $\Delta G_f^\circ(\text{H}_2\text{O}) = -237.2 \text{ kJ mol}^{-1}$, determine $\Delta G_f^\circ [\text{Al}(\text{OH})_4^-]$.

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LIST OF IMPORTANT FORMULAS

For Galvanic Cell :

$$E_{\text{cell}} = (E_{\text{oxi}})_{\text{anode}} + (E_{\text{red}})_{\text{cathode}} = (E_{\text{red}})_{\text{cathode}} - (E_{\text{red}})_{\text{anode}}$$
$$\Delta G^{\circ} = -RT \ln K = -nFE^{\circ}$$

Nernst Equation :



$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{RT}{nF} \ln \frac{[B]}{[A]}$$

Constant :

$$\frac{2.303 \times 8.314 \times 298}{96500} = 0.591$$

Thermodynamic Treatment :

$$\Delta H = -nFE + nFT \left(\frac{\partial E}{\partial T} \right)_P$$

$$\Delta S = -nF \left(\frac{\partial E}{\partial T} \right)_P$$

Faraday's Law :

$$w = zit$$

$Z \rightarrow$ electrochemical equivalent

$$Z = \frac{E}{96500}$$

$E =$ Equivaent Mass

if electrodes are in series then ratio of weight deposited will be

$$E_1 : E_2 : E_3$$

Electrolytic conductance

$$K = G \frac{\rho}{A}$$

$K \rightarrow$ specific conductance

$G \rightarrow$ conductance

$$\frac{\ell}{A} \rightarrow \text{Cell constant}$$

$$\Lambda_m = \frac{K(\text{s m}^{-1})}{1000 \times M} = \frac{1000 \times K(\text{Scm}^{-1})}{M}$$

$$\Lambda_{\text{eq}} = \frac{K(\text{Sm}^{-1})}{1000 \times N} = \frac{1000 \times K(\text{Scm}^{-1})}{N}$$

Variation of molar conductivity with concentration :

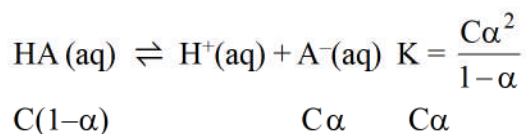
$$\Lambda_m = \lambda^{\circ}_m - A\sqrt{C}$$

$A \rightarrow$ constant



Application of Kohlrausch law :

$$\alpha = \frac{\lambda_m}{\lambda_m^0}$$

For weak electrolyte

$$K = \frac{C \times \left(\frac{\lambda_m}{\lambda_m^0} \right)^2}{1 - \frac{\lambda_m}{\lambda_m^0}}$$

For sparingly soluble salts :

$$\lambda_m(\text{AB}) = \frac{K_{\text{AB}} \times 1000}{S} = \frac{(K_{\text{sol}} - K_{\text{H}_2\text{O}}) \times 1000}{S}$$

Conductometric Titration :

Conductivity of $\text{H}^+(\text{aq})$ & $\text{OH}^-(\text{aq})$ ions has exceptionally high.

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ANSWER KEY

Q.1 A	Q.2 C	Q.3 B	Q.4 C	Q.5 B
Q.6 D	Q.7 A	Q.8 C	Q.9 C	Q.10 B
Q.11 C	Q.12 B	Q.13 B	Q.14 A	Q.15 C
Q.16 B	Q.17 B	Q.18 C	Q.19 A	Q.20 BC
Q.21 (A) P, Q (B) P, Q (C) Q, R, (D) P, S	Q.22 1.39	Q.23 43.456		
Q.24 0.372 V, -0.307 V	Q.25 0.522 V, 0.613 V			
Q.26 121.6gm, 76gm	Q.27 270	Q.28 0394	Q.29 6	
Q.30 $-1.30 \times 10^3 \text{ kJ mol}^{-1}$				

